



AMRITA
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DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

Klebsiella pneumoniae Carbapenem Resistance Prediction

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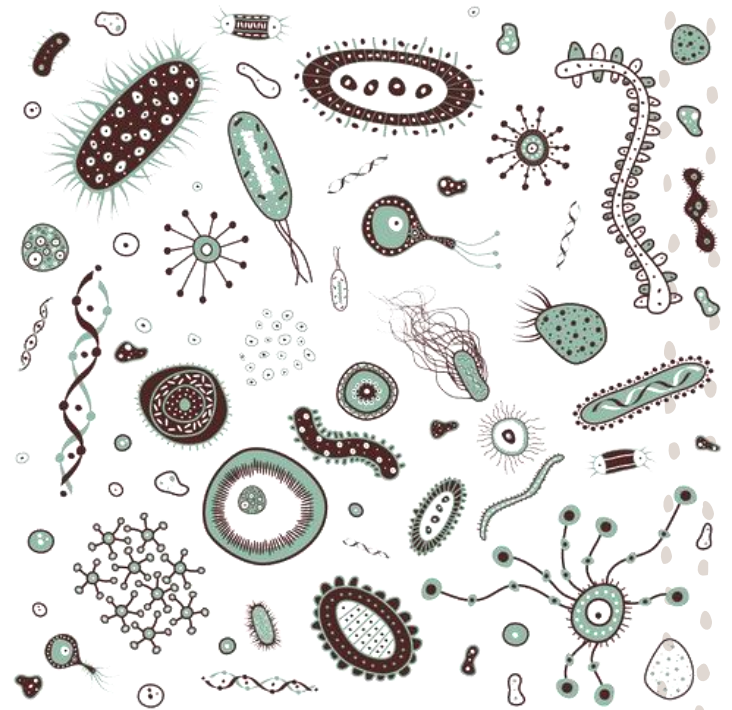
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Introduction

- Antimicrobial resistance is increasing globally
- *Klebsiella pneumoniae* causes hospital infections
- Carbapenem antibiotics are last-resort drugs
- Traditional testing takes 24–72 hours
- Machine learning can speed up prediction



Problem Statement

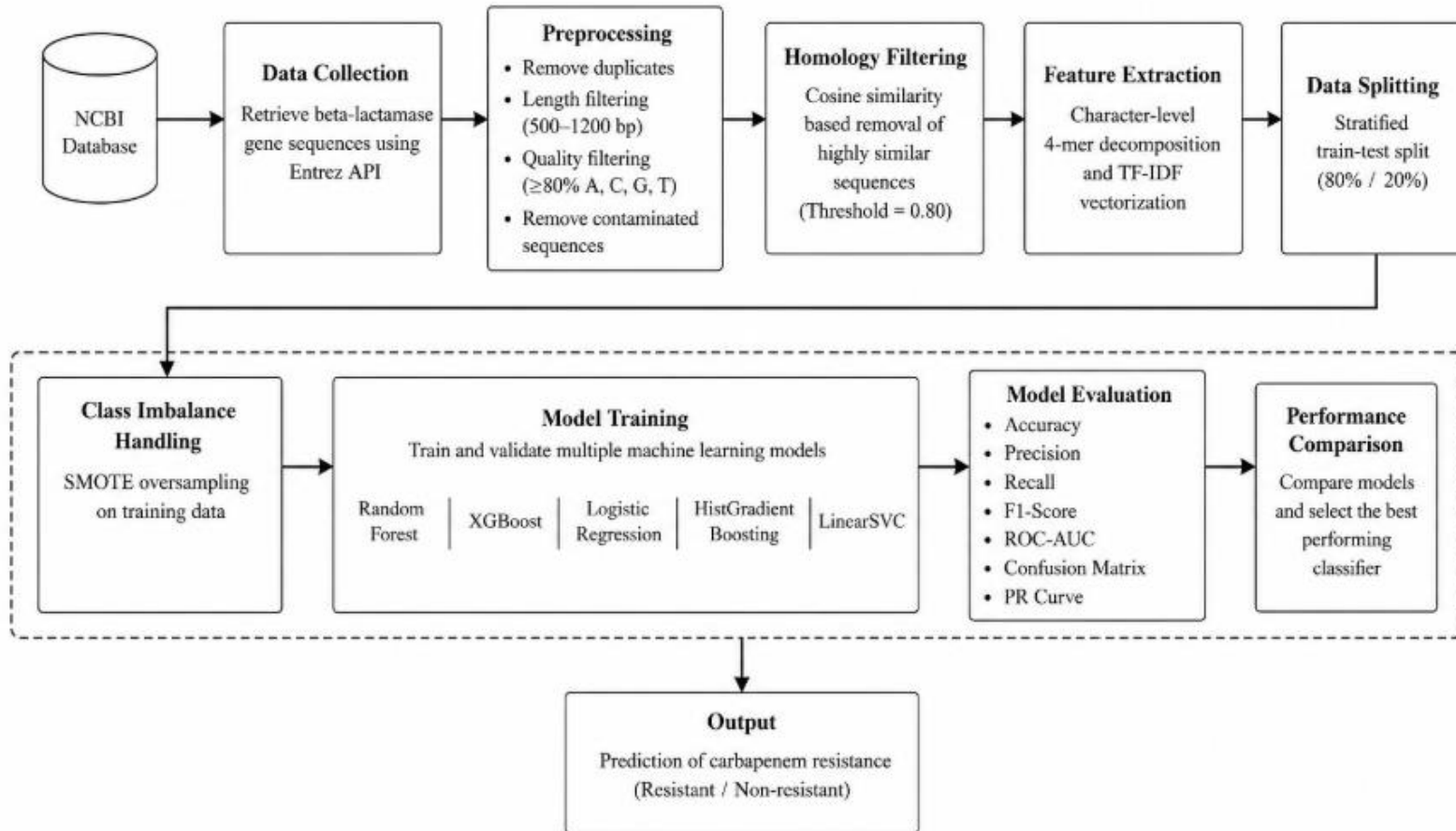
- Traditional diagnosis is slow
- Delayed treatment can lead to inappropriate treatment
- Genomic datasets contain duplicate sequences
- Repeated data can affect prediction quality
- Need for reliable prediction system

Objectives

- Collect DNA sequences from NCBI
- Clean and preprocess data
- Remove duplicate/similar sequences
- Preprocess - Sequence length- 500-1200
- ACGT->80% removes blurry data, mis-labelled data removed
- Extract DNA patterns using 4-mers
- Train ML models
- Compare model performance

System Workflow

Block Diagram of the Proposed System



Dataset Collection

Resistant genes:

- blaKPC
- blaNDM
- blaOXA-48
- blaVIM-IMP

Non-resistant genes:

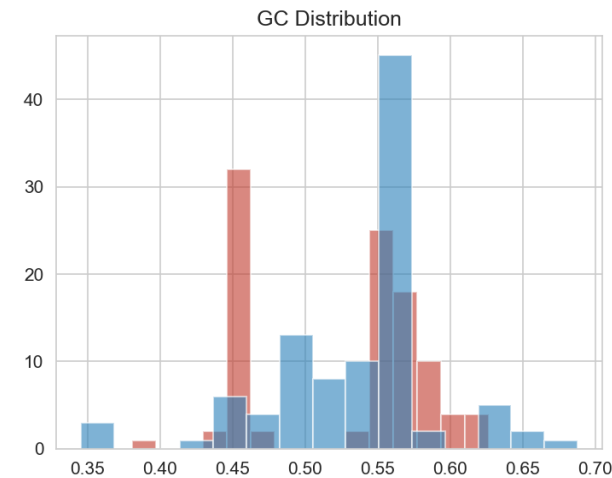
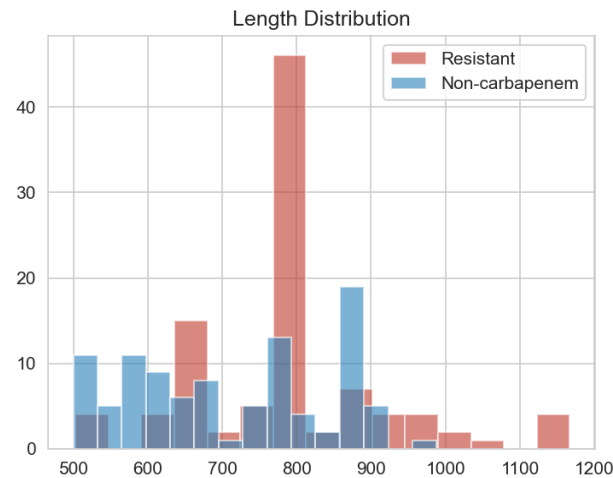
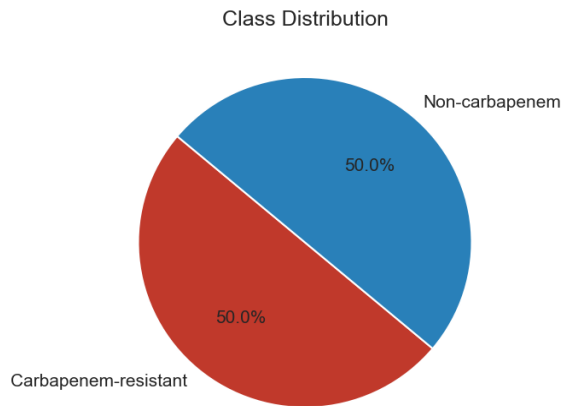
- blaTEM
- blaSHV
- blaCTX-M
- blaOXA-1

Sequences of length 1000+ were collected using the NCBI database.

Sequences after filtering were of length 1000.

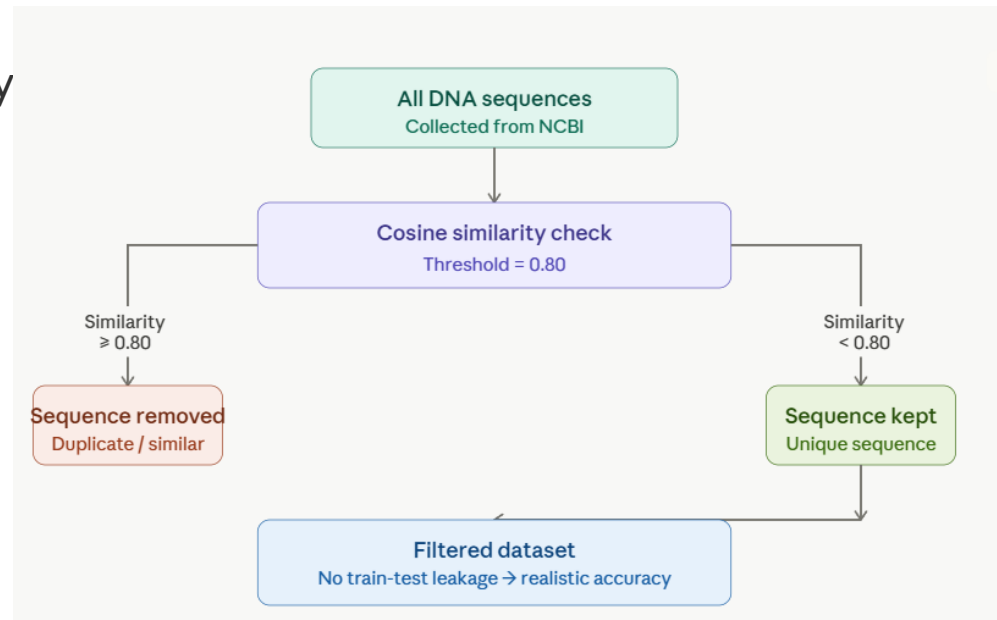
Preprocessing

- Removed duplicate sequences
- Removed low-quality DNA
- Filtered unwanted records
- Balanced dataset
- Length matching- (ex: KP-2-train and KP-3-test -> 100% acc)
- TF-IDF



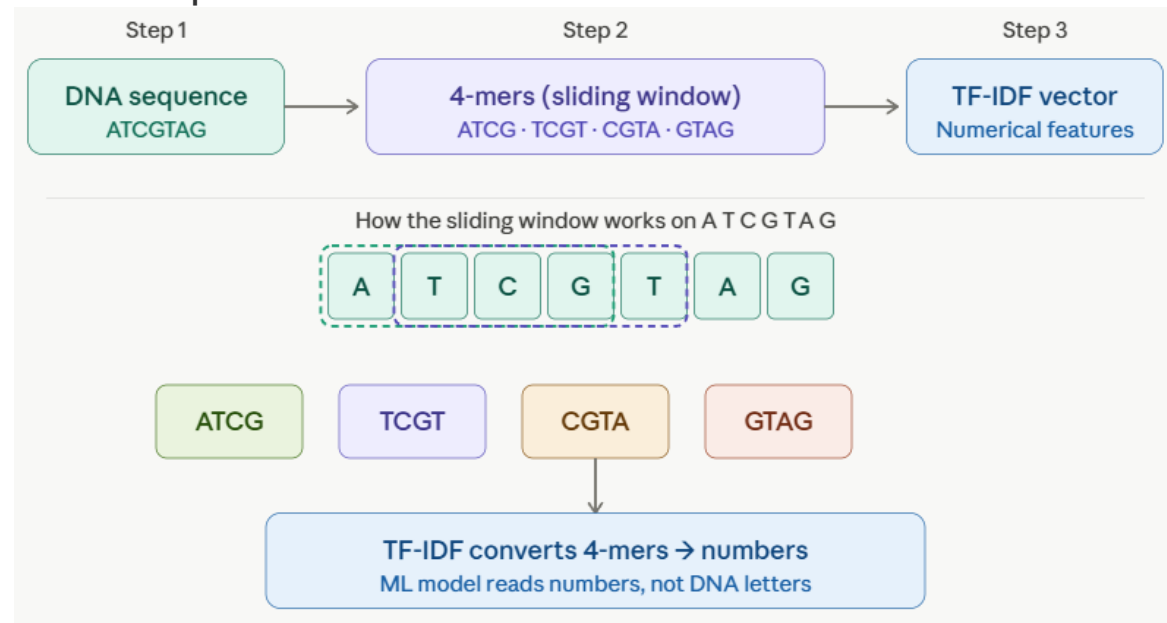
Similarity Removal

- Similar DNA sequences were identified
- Repeated patterns were removed
- Threshold set to 0.80
- Improved model accuracy



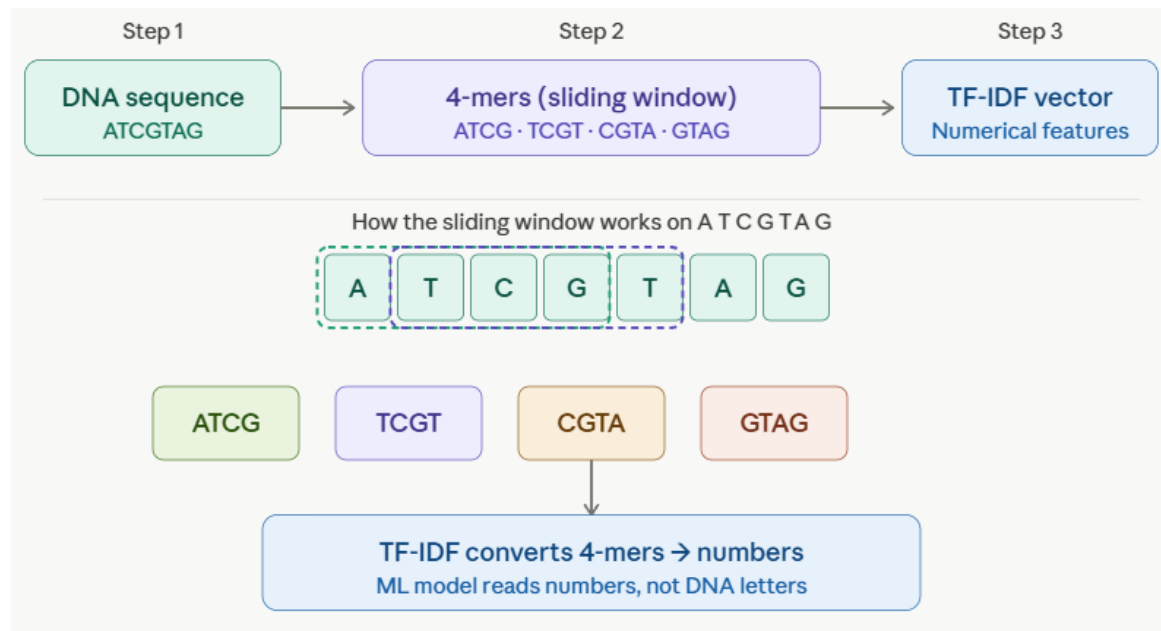
Feature Engineering

- DNA Example: ATCGTAG
- 4-mers: ATCG, TCGT, CGTA, GTAG
- DNA converted into numerical features
- TF-IDF converted DNA patterns into numerical values



Feature Engineering

- DNA sequences cannot be directly understood by ML models
- 4-mers break DNA into smaller patterns
- TF-IDF converts patterns into numerical values
- Numerical features are used for prediction



Machine Learning Models

Model	Purpose
Logistic Regression	Linear model
LinearSVC	Classification model
Random Forest	Ensemble model
XGBoost	Boosting model
HistGradientBoosting	Boosting model

Why These Models Were Selected



Model	Why Used?
Logistic Regression	To evaluate whether resistant and non-resistant DNA patterns could be separated using simple numerical feature relationships
LinearSVC	Works effectively with high-dimensional TF-IDF DNA feature representations similar to text classification data
Random Forest	Handles complex non-linear genomic feature relationships using multiple decision trees
XGBoost	Improves prediction performance by sequentially correcting classification errors
HistGradientBoosting	Efficient boosting-based approach for learning structured numerical genomic features

Literature Survey

Paper	Contribution	Output	How is it related
<i>DeepARG: A Deep Learning Approach for Predicting Antibiotic Resistance Genes from Metagenomic Data</i> - Arango-Argoty et al. (2018)	Introduced deep learning for antibiotic resistance gene prediction	Predicted AMR genes from genomic sequence data	Related to genomic AMR prediction using AI/ML techniques
<i>Colonization, Infection, and the Accessory Genome of <i>Klebsiella pneumoniae</i></i> - Martin & Bachman (2018)	Explained genomic structure and resistance mechanisms of <i>Klebsiella pneumoniae</i>	Identified clinically important resistance-associated genomic features	Related to biological importance of <i>Klebsiella pneumoniae</i> resistance

Literature Survey

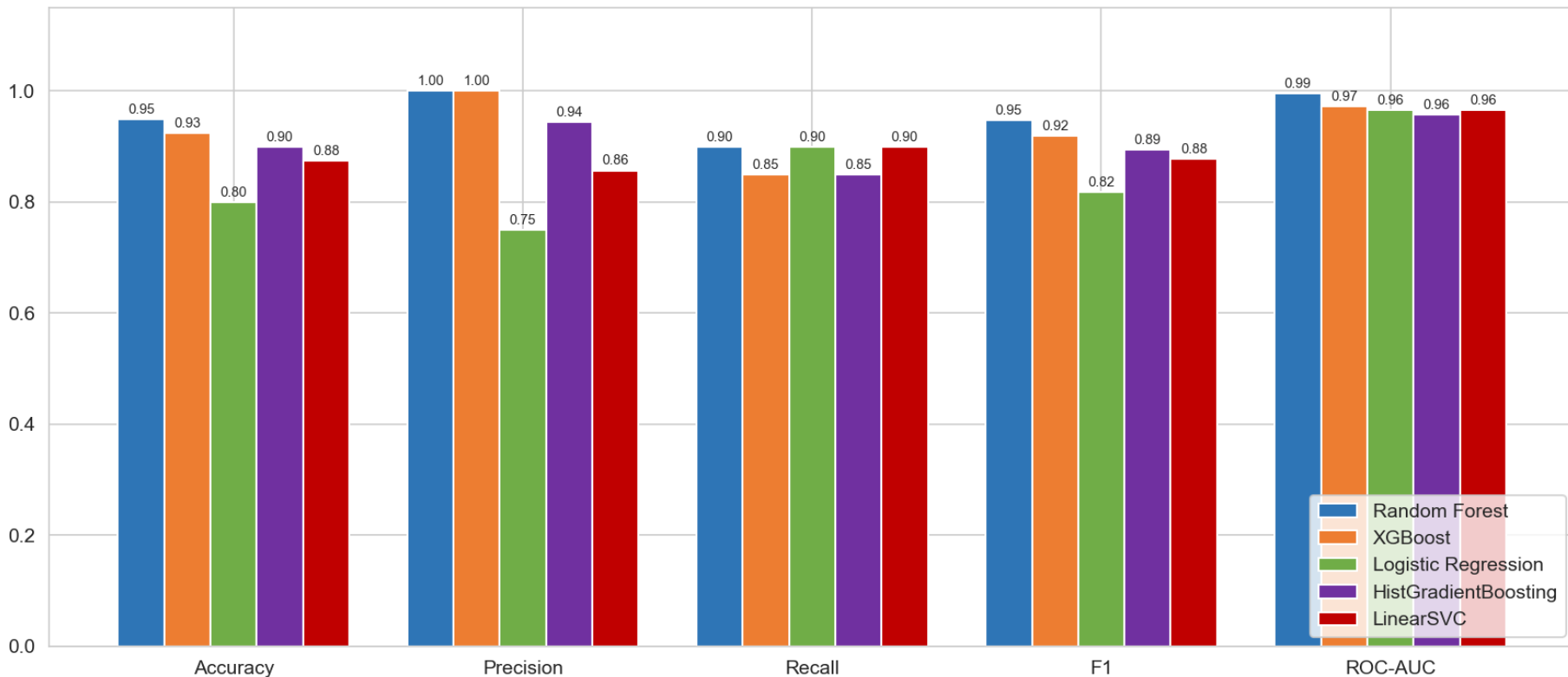
Paper	Contribution	Output	How is it related
<i>Genome-Based Prediction of Bacterial Antibiotic Resistance</i> – Su et al. (2019)	Applied machine learning for genome-based bacterial resistance prediction	Predicted antibiotic resistance using genomic features	Related to ML-based genomic AMR prediction workflow
<i>Machine Learning for Classifying Tuberculosis Drug-Resistance from DNA Sequencing Data</i> – Yang et al. (2018)	Used ML models for drug resistance classification from DNA sequences	Classified resistance using DNA sequencing features	Related to DNA sequence-based machine learning classification

Evaluation Metrics

Metric	Meaning
Accuracy	Overall correctness
Precision	Correct resistant predictions
Recall	Resistant detection ability
F1 - Score	Balance of precision and recall
ROC - AUC	Overall classification quality

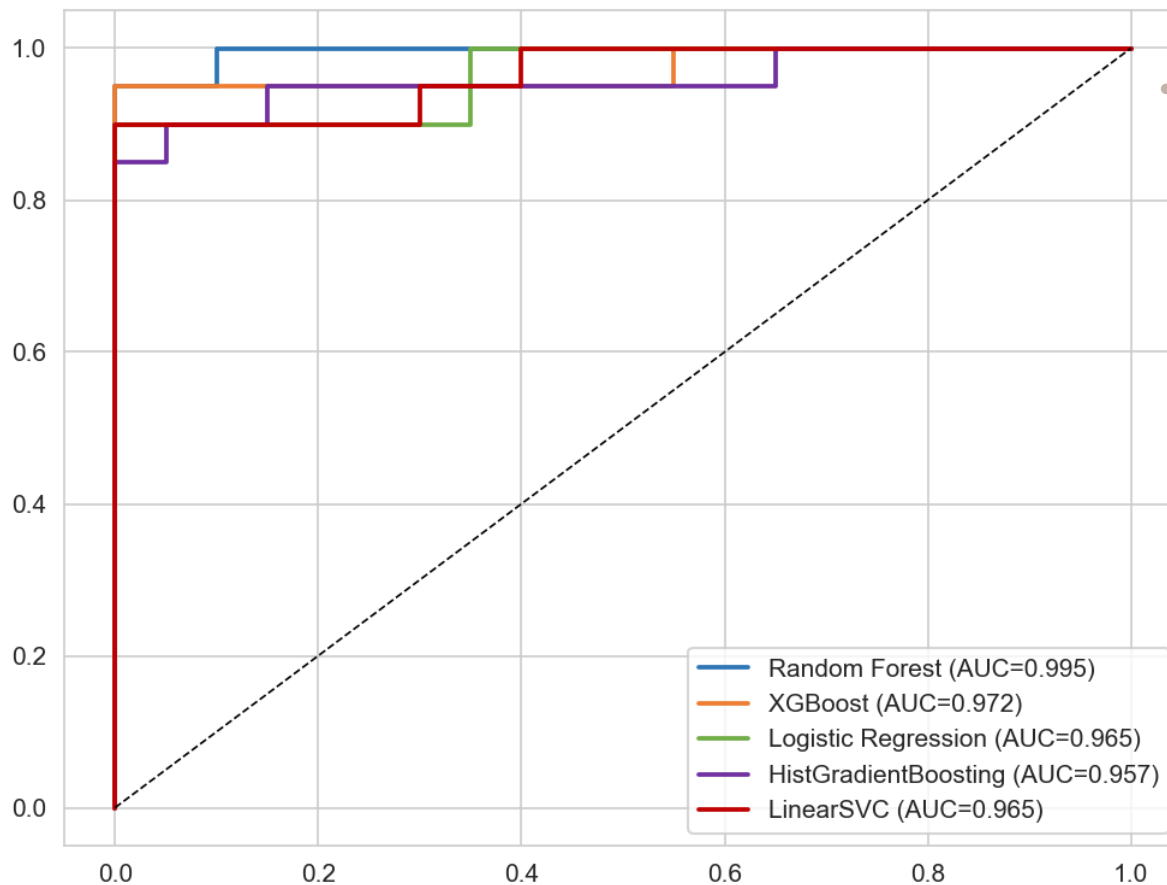
Results

- Accuracy ranged between 88% and 95%.
- Random Forest showed best performance
- Models classified resistant samples effectively
- ROC curves showed strong prediction ability



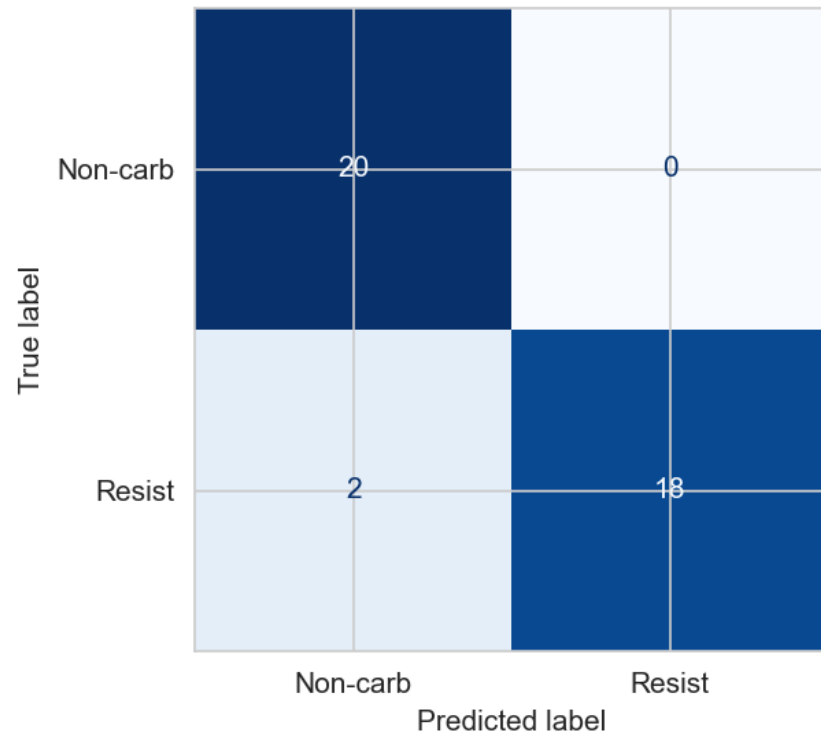
ROC Curve Analysis

- ROC curve evaluates prediction performance
- Curves closer to top-left indicate better classification
- Random Forest achieved highest ROC-AUC



Confusion Matrix – Random Forest

- Most samples classified correctly
- Very few incorrect predictions
- Resistant samples identified effectively



Applications

- Hospital diagnostics
- Faster AMR screening
- Genomic surveillance
- Precision medicine



Future Scope

- Deep learning models
- DNABERT
- Real-time hospital systems
- Multi-drug prediction

Conclusion

- Developed ML-based AMR prediction system
- Removed repeated DNA patterns
- TF-IDF 4-mers improved prediction
- Random Forest gave best results
- Useful for rapid genomic screening

Novelty

- Uses different resistant(*bla*KPC, *bla*NDM, *bla*OXA-48) and non-resistant(*bla*TEM, *bla*SHV, *bla*CTX-M) gene families for better prediction.
- Applies SMOTE after feature extraction to balance resistant and non-resistant samples before model training
- Performs dataset checking before training to identify repeated or biased DNA sequence patterns

Thank You

